NULL-TECH
CIRCUIT STUDIO

Technical & business proposal

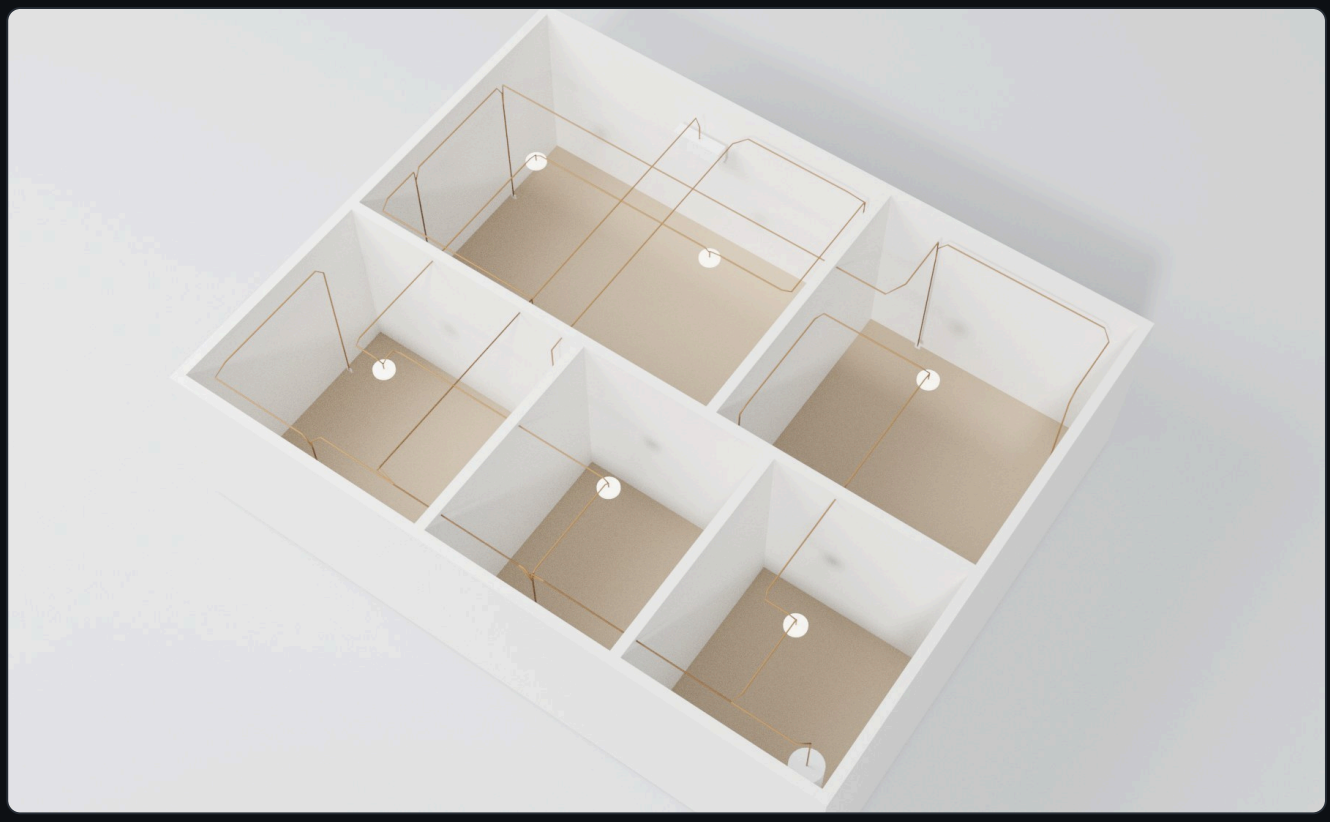
An electrical-engineering workspace to design, simulate, calculate, visualise in 3D and document electronic circuits, building installations and control systems — operated by mouse and keyboard or by hand gestures through a webcam.

59
component types in 16 categories

3
engineering modes: electronic · installation · control

153
automated tests passing + 5 end-to-end workflows

0
servers required — runs offline on the user's computer



Version 0.2 · cover: Blender render of the sample flat exported from the software

November 2026

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1 Executive summary

NULL-TECH Circuit Studio already exists as working software. A user can start from an empty workspace, build an electronic circuit, a building electrical installation or a relay/PLC control circuit, simulate or calculate it, inspect it in 3D and print engineering drawings — all generated from one engineering model.

One model, every view

2D, 3D, simulation, calculations, drawings and exports derive from the same data and cannot disagree.

Verified physics

Simulator results agree with ngspice (the reference open-source SPICE engine) within 0.1 mV on every DC example.

Touch-free control

Webcam gestures use exactly the same tools as the mouse, with preview and confirmation before any change.

What we propose

1. **Pilot (3 months):** deploy in university laboratories, run an independent review of every calculation rule by a licensed electrical engineer, and test hand tracking with real webcams and users.
2. **Productisation (≈ 9–12 months):** complete the safety calculations, localise (Arabic, Kurdish Sorani; regional rules), and ship signed desktop installers.
3. **Commercial launch:** free student edition plus institutional and professional licences (options in §9.5).

Indicative effort for the full programme is **≈ 48 person-months** (§19). Running costs are low because the product needs no servers: it runs on the user's own computer.

> **Positioning.** The software assists engineers; it does not replace them. No result is ever labelled “safe” or “compliant”. Checks that are not implemented are shown as *not evaluated*, and every calculation shows its formula, inputs, assumptions and source.

Evidence tags used in this document: MEASURED obtained by running the software or its tests; ESTIMATE engineering or planning judgement; ASSUMPTION a value to be replaced with local data.

2 Problem and opportunity

Problem today	Consequence	How NULL-TECH addresses it
Fragmented tools: schematic editor, simulator, spreadsheets, CAD, 3D viewer	The same design is re-entered several times and copies drift out of date	One model feeds every view, calculation and export
Opaque calculators	A number without formula, assumptions or source is risky for safety-relevant work	Every result shows formula, inputs, assumptions and cited standard
Teaching needs hands-on construction	Students watch rather than build; lab equipment is limited	15 buildable examples, live simulation, automatic lab report
Hands-busy settings (labs, workshops, presentations)	Mouse use interrupts work and demonstrations	Webcam gesture control on the same tools
Local rules and language	International tools rarely reflect regional requirements or Arabic/Kurdish users	Rules are project data, not code; localisation layer prepared (RTL-ready)

3 The product

Mode	Capabilities in the current version
Electronic circuits	Resistors, capacitors, inductors, DC/AC/current sources, diodes, LEDs, Zeners, BJTs, MOSFETs, op-amps, logic gates, relays, transformer, connectors, probes. Wiring with snapping and junctions, net labels. DC operating point, transient waveforms, AC frequency response. Rating checks (power, current, voltage, polarity).
Installations	Rooms (drawn or imported from DXF), distribution boards, luminaires, switches, sockets, appliances, cable trays. Circuits created from a board; cables routed through trays; lengths from the actual route. Load, design current, voltage drop, suggested breaker and conductor size, earth-fault loop impedance (TN/TT), breaking capacity, phase balance, schedule, BOM, single-line diagram.
Control systems	Push buttons, E-stop, limit switch, overload contact, contactor coils with tagged contacts, pilot lamps, breakers, PLC inputs/outputs and a PLC program in IEC 61131-3 Structured Text, all executed by the simulator.

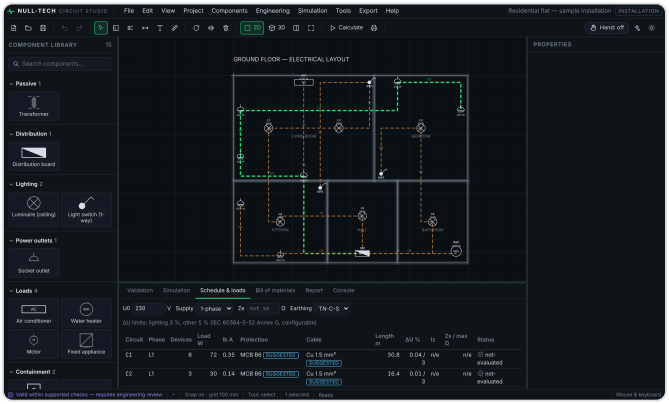


Fig. 1 — Installation plan: circuit C3 highlighted; schedule with Ib, protection, cable, length, ΔU, Zs and status.

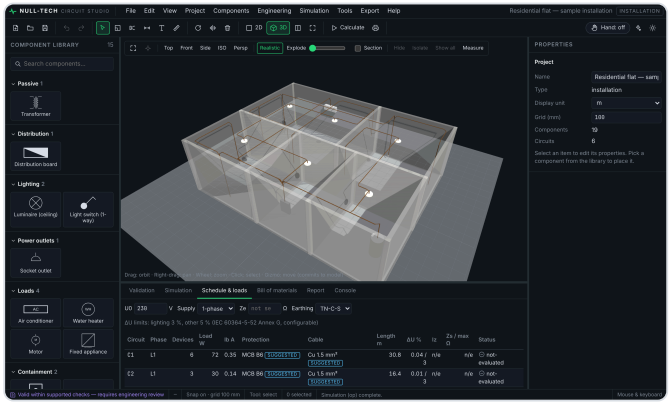


Fig. 2 — Same model in 3D: walls, devices at mounting height, cables at route height.

Cross-cutting capabilities

Area	Capability
3D	Parametric component models with terminals exactly on schematic pins; orbit, orthographic/perspective, Top/Front/Side/ISO, section plane, exploded view, hide/isolate, measuring; LED and lamp glow from simulated current; moving a part in 3D edits the model.
Drawings	A0–A4 and custom sheets, scale 1:N or fit, ISO frame with zone references, title block, revisions, schematics, installation/lighting/power plans, single-line diagram, schedules, BOM; vector PDF, SVG, PNG, JPEG, browser printing.
Exchange	Native `.ntcs.json` (versioned, validated, migrated, JSON Schema); DXF R12 export/import; SPICE export/import; GLB, glTF, OBJ, STL.
Input	Mouse, keyboard (customisable shortcuts), hand tracking; preview → validate → confirm/reject; undo/redo for every operation.
Education	15 examples (Ohm's law to motor starters and PLC logic) with theory, hand calculation and steps; engineering/lab report export.
Assistant	Offline rule-based assistant: “create a red LED circuit with 12 V”, “place 4 lights in living room”, “why is this not working”. Proposals are previewed and validated like any other operation.
Platforms	Any modern browser (served locally), or a Tauri desktop application with native save/open dialogs.

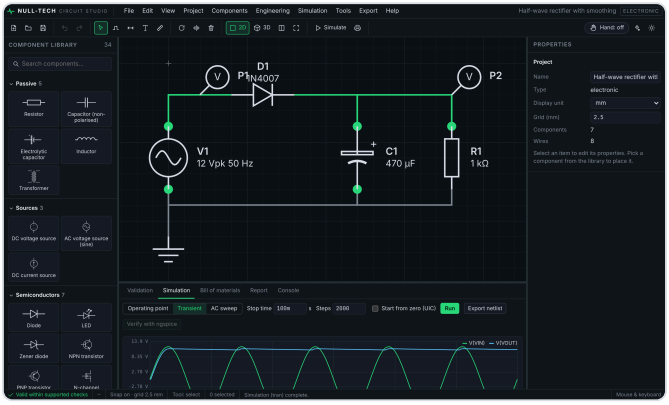


Fig. 3 — Half-wave rectifier transient: input sine and smoothed output.

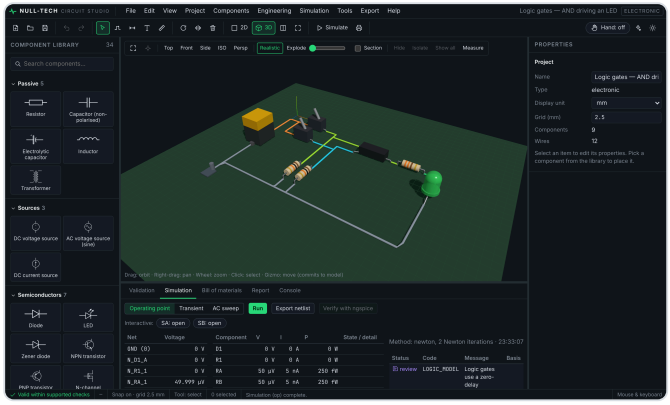


Fig. 4 — Logic-gate circuit in 3D; gate, resistors and LED placed at schematic positions.

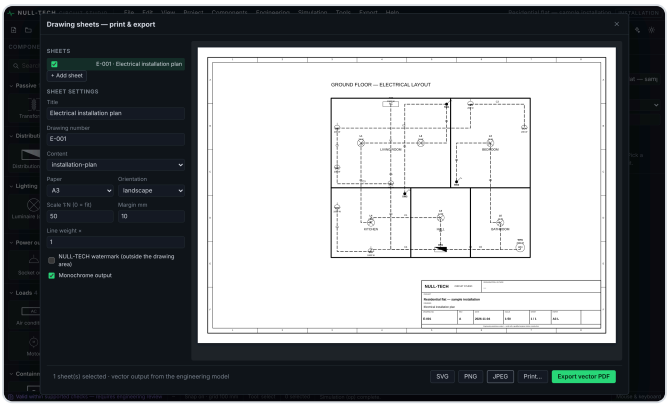


Fig. 5 — Drawing sheet with ISO frame and title block (vector PDF export).

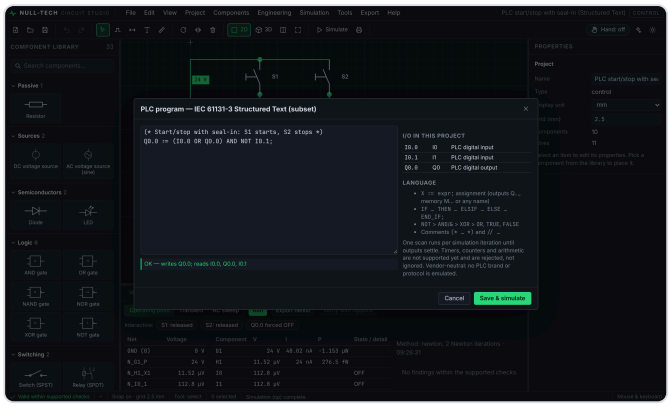


Fig. 6 — PLC program (Structured Text) driving the simulated control circuit.

4 Users, personas and journeys

Four primary user groups were derived from the product brief. Each journey below is a workflow the current version already supports end to end.

Persona	Goals	Pain points today	What NULL-TECH gives them
Engineering student 2nd–4th year, electrical / electronics	understand circuits, pass labs, write reports, prepare for practice	theory feels abstract; limited lab time; separate tools for drawing and simulating	build circuits by hand, see voltages live, compare with hand calculations, export a lab report
Instructor / lab supervisor	teach concepts clearly, assess many students, run demonstrations	preparing material; checking students' work; equipment wear	15 ready examples with theory; consistent results; gesture control at the projector
Electrical designer residential / commercial	lay out installations fast, size cables and breakers, issue drawings	spreadsheets for loads and ΔU disconnected from the drawing; re-entering data	plan → circuits → calculations → schedule → drawings from one model, with every assumption visible
Technician / trainee training centres	practise control wiring and PLC logic safely	no access to real panels; risk when experimenting	motor-starter, relay and PLC circuits that respond to button presses in simulation

Journey 1 — student lab: LED resistor sizing

#	Step	What the software does
1	Help ▸ Examples ▸ "LED with series resistor" (or build it)	loads an editable project; theory shows I ≈ (9 − 1.9)/470
2	Press F5	solves the circuit in 0.4 ms; shows 9 V and 1.89 V on the wires
3	Change R1 to 100 Ω	LED current exceeds 20 mA → `ERROR` with the rule that triggered it
4	Open 3D	LED glows in proportion to the simulated current
5	Report tab ▸ Download	lab report with net voltages, currents, power, method and findings

Journey 2 — designer: residential installation

#	Step	What the software does
1	Import the architect's DXF plan as rooms	closed polylines become rooms (metres converted to millimetres)
2	Place board, luminaires, sockets, cable trays	devices get default powers flagged as *assumed*
3	Circuit tool: board → devices → Enter	cables routed through trays; preview with Ib, length and ΔU before confirming
4	Enter supply Ze and earthing system	Zs and disconnection checks per circuit; breaking capacity
5	Print ▸ plan, single-line, schedule, BOM	multi-sheet vector PDF with title blocks

Journey 3 — instructor demonstration with gestures

Instructor stands at the projector: points to move the cursor, pinches on a pin and drags to draw a wire, makes a fist to confirm the preview, a V-sign to reject it, opens the palm to pause. No mouse needed; any mistake can be undone.

5 Physics and engineering foundations

This section lists the physical laws and numerical methods the software actually implements (source: `frontend/src/simulation`, `frontend/src/calculations`).

5.1 Circuit laws and network formulation

Ohm's law, Kirchhoff's current and voltage laws are enforced through **Modified Nodal Analysis** (MNA). For n nodes and m voltage-defined branches:

```
[ G   B ] [ v ]   [ i ]      G : conductances (n×n)       v : node voltages
[ C   D ] [ j ] = [ e ]      B,C : branch incidence       j : branch currents (sources, inductors,
op-amp outputs)
                             i : injected currents        e : source voltages
```

Solved by LU decomposition with partial pivoting. A 1 pS shunt on every node keeps floating nodes solvable; nets with no DC path to ground are reported as warnings.

5.2 Non-linear devices — Newton–Raphson

```
x(k+1) = x(k) − J(x(k))⁻¹ · F(x(k))     converged when |Δv| ≤ 10⁻⁴·|v| + 1 µV
fallbacks: gmin stepping (10⁻² → 10⁻¹² S), then source stepping (5 % → 100 %)
```

Device	Model equation
Diode / LED	`I = Is·(e^(V/(N·Vt)) − 1)`, series Rs; LED Is fitted so I = 20 mA at the stated Vf (N = 2); SPICE junction-voltage limiting (pnjlim)
Zener	forward Shockley term + breakdown `−IBV·e^(−(V+BV)/(N·Vt))`
Thermal voltage	`Vt = k·T/q = 1.380649e−23 × 300.15 / 1.602176634e−19 = 25.865 mV` (SPICE default 27 °C)
BJT	Ebers–Moll transport: `Ic = Is(e^(Vbe/Vt) − e^(Vbc/Vt)) − (Is/βR)(e^(Vbc/Vt) − 1)`, `Ib = (Is/βF)(e^(Vbe/Vt) − 1) + (Is/βR)(e^(Vbc/Vt) − 1)`
MOSFET	Shichman–Hodges level 1: saturation `Id = K/2·(Vgs − Vth)²·(1 + λVds)`; linear `Id = K·((Vgs − Vth)Vds − Vds²/2)·(1 + λVds)`
Op-amp	rail-limited high-gain source `Vout = Vmid + Vhalf·tanh(A·Vd / Vhalf)`
Transformer	ideal: `Vp = n·Vs`, `Is = −n·Ip` (power conserving)
Relay / contactor	coil = resistance; contacts change at pick-up / drop-out thresholds; iterated to a consistent state; oscillation detected
Logic gate	input HIGH if V ≥ VIH, LOW if V ≤ VIL (CMOS: 0.7/0.3 × Vcc; 74LS: 2.0/0.8 V); output VOH/VOL through Rout; zero delay
PLC	Structured Text scan each iteration: inputs from node voltage ≥ threshold, outputs switch a sourcing supply onto the terminal

5.3 Energy-storage elements — transient analysis

```
Capacitor  i = C·dv/dt   trapezoidal companion:  i(n+1) = (2C/h)·v(n+1) − [(2C/h)·v(n) + i(n)]
Inductor   v = L·di/dt   trapezoidal companion:  v(n+1) = (2L/h)·[i(n+1) − i(n)] − v(n)
First step after t = 0 uses backward Euler to avoid trapezoidal start-up error.
```

5.4 Frequency domain — AC analysis

```
Linearised at the operating point; capacitor admittance Y = jωC; inductor impedance Z = jωL;
complex MNA solved per frequency on a logarithmic sweep → magnitude (dB) and phase (°).
```

5.5 Building-installation physics

Quantity	Formula implemented	Source
Apparent power, demand	`S = P / cos φ`; `S_d = Σ S × demand factor` (arithmetic sum, conservative)	basic AC power
Design current	single-phase `Ib = S_d / U0`; three-phase `Ib = S_d / (3·U0)`	basic AC power
Conductor resistivity	ρ20 Cu = 0.017241 Ω·mm²/m; Al = 0.028264; in service `ρ1 = 1.25 × ρ20`	IEC 60028; IEC 60364-5-52 Annex G
Cable length	plan length of the drawn route + vertical drops (route height ↔ mounting heights)	model geometry
Voltage drop	`u = b·(ρ1·L/S·cos φ + λ·L·sin φ)·I`, λ = 0.08 mΩ/m, b = 2 (1-ph) or 1 (3-ph), per segment with downstream current; `ΔU% = 100·Σu / U0`	IEC 60364-5-52 Annex G
Breaker suggestion	smallest rated current In ≥ Ib from 6, 8, 10, 13, 16, 20, 25, 32, 40, 50, 63, 80, 100, 125 A	IEC 60898-1
Conductor suggestion	smallest IEC 60228 size ≥ 1.5 mm² (Cu) meeting the ΔU limit (3 % lighting / 5 % other, configurable)	IEC 60228; IEC 60364-5-52
Earth-fault loop	`Zs = Ze + L·ρ1·(1/S + 1/S_PE)`; TN condition `Zs·Ia ≤ U0`, Ia = 5/10/20 × In for curve B/C/D	IEC 60364-4-41 411.4.4; IEC 60898-1 Table 7
TT systems	`RA·IΔn ≤ 50 V` and an RCD present	IEC 60364-4-41 411.5.3
Breaking capacity	prospective fault current at the board `U0 / Ze ≤ Icn`	IEC 60364-4-43 434.5.1
Current-carrying capacity	In ≤ Iz only from rows the designer enters with their source	user-supplied (no table invented)

5.6 Engineering disciplines involved

Discipline	Where it appears in the product
Circuit theory & electronics	MNA solver, device models, rating checks, examples
Power & installation engineering	load, cable sizing, voltage drop, protection, fault loop, boards, schedules
Control & automation	relay logic, contactors, PLC I/O, Structured Text interpreter
Numerical methods	LU decomposition, Newton–Raphson, continuation, implicit integration, complex solving
Computer graphics	Three.js scene, parametric meshes, incremental sync, clipping, picking
Computer vision & signal processing	MediaPipe hand landmarks, gesture features, One Euro filter
Graph algorithms	union-find net extraction, Dijkstra routing through cable trays
Technical drawing	ISO sheet sizes, frames, title blocks, scales, vector output
Software engineering	immutable model, command history, schema migration, automated testing, CI

6 Validation against independent references

Every value below comes from an automated test that runs on each build MEASURED.

Case	Independent reference	Expected	Software	Tolerance
Ohm's law, 9 V / 1 kΩ	I = V/R	9.000 mA	9.000 mA	10^{-9}
Two-source network, node A	KCL by hand: 75/11 V	6.8182 V	6.8182 V	10^{-6}
Voltage divider 10 k / 4.7 k from 10 V	Vin·R2/(R1+R2)	3.1973 V	3.1973 V	10^{-6}
RC charging at t = τ = 0.1 s	$5\cdot(1 - e^{-1})$	3.1606 V	3.161 V	5×10^{-4}
RC low-pass at fc	$-20\cdot\log_{10}\sqrt{2}$, −45°	−3.010 dB	−3.010 dB	10^{-3}
Ideal transformer 10:1, 23 Ω load	power balance 23 W / 230 V	0.100 A	0.100 A	10^{-6}
All DC examples (11 circuits)	**ngspice-42** operating point	every node agrees		0.1 mV + 0.01 %
Rectifier and op-amp transients	**ngspice-42** transient	waveforms agree		2 % / 0.01 V
Voltage drop, 2 kW, 20 m, 2.5 mm² Cu	Annex G by hand	2.998 V	2.998 V	10^{-3}
R1+R2, 20 m, 2.5 / 1.5 mm²	L·ρ1·(1/S + 1/S_PE)	0.4598 Ω	0.4598 Ω	10^{-9}
Max Zs, B16 breaker, 230 V	U0 / (5·In)	2.875 Ω	2.875 Ω	10^{-9}
Cable routed through trays	sum of route segments	10 000 mm	10 000 mm	10^{-6}

Behaviour checks (pass / fail)

- LED over-current, resistor over-power, capacitor reverse polarity → reported as ERROR, never silently accepted.
- Missing ground, voltage-source loops, PLC syntax errors → simulation refused with a precise message.
- Inverter ring oscillator → oscillation detected instead of a fake steady state.
- Motor-starter seal-in, relay pick-up, PLC start/stop, AND-gate truth table → correct states in every combination.
- Without supply impedance Ze, the fault-loop check is NOT EVALUATED — never passed.

Performance on the development machine MEASURED

Operation	Time	Operation	Time
DC operating point, LED circuit	0.37 ms	Installation calculation, 6-circuit flat	0.22 ms
Transient, rectifier, 2 000 steps	11.6 ms	Vector PDF of one A3 sheet	1.7 ms
AC sweep, 1 Hz–1 MHz, 121 points	2.0 ms	DXF export of the flat	0.32 ms
Gesture recognition + commands, per frame	0.011 ms	Main application bundle (gzip)	≈ 0.33 MB

Timings are from a cloud development container; they show the calculations are far below interactive limits, not a guarantee for every device. Webcam landmark detection time depends on the camera and CPU and has not yet been measured on real hardware.

7 Worked engineering examples

The following numbers are the software's own output for the shipped examples MEASURED, followed by an independent hand check.

Example A — LED with series resistor (9 V, 470 Ω, red LED Vf = 1.9 V at 20 mA)

Quantity	Software	Hand check
LED current	15.137 mA	approximation (9 − 1.9)/470 = 15.1 mA; exact value solves 9 = 470·I + 2·Vt·ln(I/Is)
LED voltage	1.8856 V	slightly below 1.9 V because I < 20 mA — consistent with the diode law
Resistor voltage / power	7.1144 V / 107.7 mW	7.1144 × 0.015137 = 0.1077 W (43 % of a 0.25 W rating — below the 50 % margin)
Source power delivered	136.2 mW	9 V × 15.137 mA = 136.2 mW = 107.7 + 28.5 mW (energy balance)

Example B — residential flat, final circuits (U0 = 230 V, Ze = 0.35 Ω entered, Cu PVC)

Circuit	Devices	Load W	cos φ	Ib A	Route m	ΔU %	S mm²	In A	R1+R2 Ω	Zs / max Ω
C2 lighting	3	30	0.90	0.145	16.4	0.015	1.5	B6	0.471	0.821 / 7.667
C3 sockets	5	1 000	0.90	4.831	47.9	1.533	1.5	B6	1.376	1.726 / 7.667
C4 kitchen	2	2 000	0.90	9.662	16.7	1.324	1.5	B10	0.480	0.830 / 4.600
C5 air-con	1	1 500	0.85	7.673	11.9	0.973	1.5	B8	0.342	0.692 / 5.750
C6 water heater	1	2 000	1.00	8.696	6.6	0.717	1.5	B10	0.190	0.540 / 4.600

Board DB1 total demand 7 211 VA (31.4 A on L1). Bill of materials: 130.3 m of Cu 3×1.5 mm² cable measured along the routes. Cable sizes and breakers are *suggestions*; current-carrying capacity is not evaluated until the designer enters a sourced ampacity table.

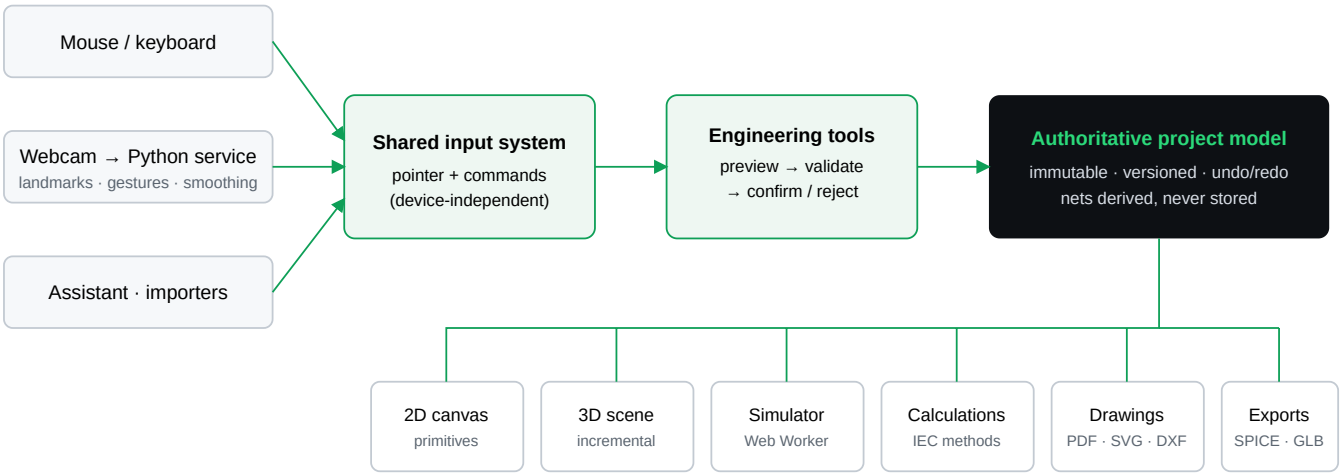
Hand check of circuit C3

```
S_demand = 1 000 W / 0.9 = 1 111.1 VA            Ib = 1 111.1 / 230 = 4.831 A   ✓
Breaker: smallest IEC 60898-1 rating ≥ 4.831 A → 6 A (curve B)   ✓
ρ1 = 1.25 × 0.017241 = 0.021551 Ω·mm²/m
R1+R2 = 47.9 m × 0.021551 × (1/1.5 + 1/1.5) = 1.376 Ω        ✓
Zs = Ze + R1+R2 = 0.35 + 1.376 = 1.726 Ω
Ia = 5 × 6 A = 30 A  →  max Zs = 230 / 30 = 7.667 Ω  →  1.726 ≤ 7.667  condition met   ✓
Voltage drop uses the downstream current in each of the 5 cable segments (daisy chain),
so the total 1.533 % is lower than the worst-case estimate 2·ρ1·L·Ib/S = 6.65 V (2.9 %).
```

Example C — control logic (DOL motor starter)

Action	K1 coil	H1 lamp	Physics / logic
Initial	off	off	no path through S1 (open)
Press S1	on	on (50 mA)	24 V across 200 Ω coil = 120 mA > pick-up; lamp 24 V / 480 Ω
Release S1	**on**	on	seal-in contact K1.1 keeps the coil energised
Press S0 (stop)	off	off	NC contact opens; coil voltage below drop-out

8 System architecture



The Python service only emits input commands over a local WebSocket (127.0.0.1, localhost origins only).
It never receives or changes the project. ngspice can be called through it for independent verification.

Decision	Reason
Web technology (TypeScript, React, Three.js) + Tauri desktop shell	One codebase for browser and desktop; WebGL for 3D; small installers using the system web view.
Python local service	Mature computer-vision stack (OpenCV, MediaPipe) and easy ngspice integration.
Built-in simulator + ngspice as reference	Works with nothing installed; ngspice provides independent verification.
Local-first, no server	Offline use, privacy, near-zero running cost.
Reference data as cited JSON	Rules can be replaced per jurisdiction without code changes.

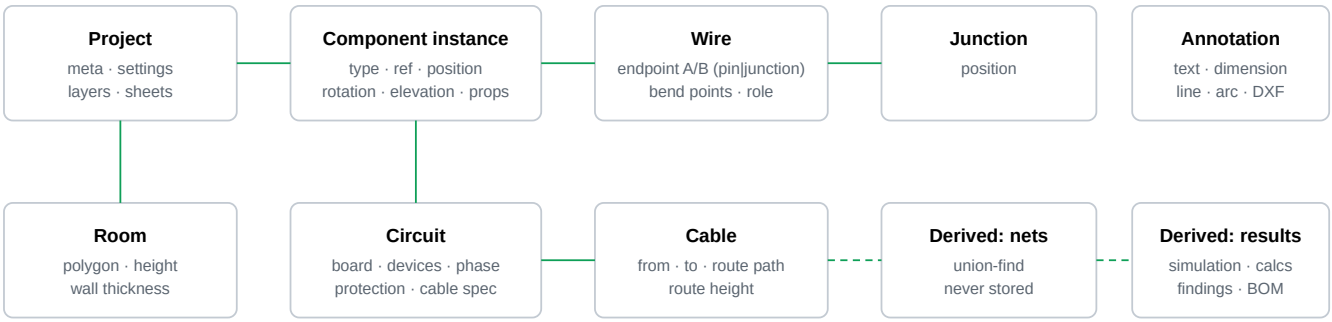
Codebase size **MEASURED**: ≈ 13 000 lines of TypeScript (application), ≈ 1 000 lines of Python (service), ≈ 1 800 lines of tests.

9 Models

9.1 Simulation models

44 of the 59 component types take part in simulation; the remaining 15 are documentation or installation devices (generic ICs, microcontrollers, boards, luminaires …) and are reported as *not evaluated* where relevant. Device equations are listed in §5.2.

9.2 Engineering data model



All lengths in millimetres; stable random IDs; JSON Schema published in `shared/schemas/project.schema.json`; schema version and step-wise migrations.

9.3 Computer-vision model

Stage	Model / method
Hand landmarks	MediaPipe Hands (pre-trained by Google, used as-is): 21 landmarks per hand, up to 2 hands.
Gesture classification	Deterministic geometric rules (joint angles, distances normalised by palm size) — no training data needed, fully explainable. Verified for all poses at three hand sizes and ±35° tilt.
Temporal filtering	State machine: N stable frames (default 4; pinch 2), pinch hysteresis 0.32 / 0.45 × palm, 0.6 s command cool-down.
Cursor smoothing	One Euro filter (Casiez et al., 2012), min cut-off 1 Hz, β = 4.

9.4 3D models

Parametric meshes for about 40 model types (axial resistor with IEC 60062 colour bands computed from the resistance value, capacitors sized by capacitance, LEDs, TO-92/TO-220, DIP packages with real pin count, relays, transformers, boards, luminaires, sockets, appliances, cable trays …). See §10.

9.5 Business model options FOR DECISION

Option	Description	Fit
Free student edition	Full electronic and control modes; installation mode for study	Adoption in universities
Institutional licence	Per-laboratory or campus licence, teaching material, support	Universities, training centres
Professional licence	Per-seat; installation drawings, schedules, regional rule packs	Design offices, contractors
Open-core	Core open source; paid rule packs, support and desktop builds	Community growth, trust in calculations

10 3D pipeline

Aspect	Specification
Coordinate system	2D plan (x, y↓) → 3D (X = x, Z = y, Y = elevation). Units millimetres. 2D rotation θ → 3D rotation about Y of −θ; mirror = scale X −1.
Correspondence	Lead tips and terminals of every 3D model sit on the library pin coordinates — verified by an automated test (within 1 mm).
Rendering	Three.js WebGL; physically based materials; hemisphere + directional light with soft shadow maps (2048^2); sRGB output.
Performance	Incremental synchronisation: each entity has a content hash, only changed entities are rebuilt; shared materials are cached.
Interaction	Orbit, pan, zoom; perspective/orthographic; Top/Front/Side/ISO; fit to project/selection; picking into the shared selection; move gizmo that writes back to the model (snapped to grid); hand-gesture orbit and selection.
Analysis views	Horizontal section plane, exploded view, hide/isolate/show all, point-to-point measuring, simplified vs realistic materials.
Simulation feedback	LEDs glow in proportion to simulated current; pilot lamps and relay indicators light when energised.
Installation scenes	Floors and walls extruded from room polygons (shared walls de-duplicated); devices at mounting height; cables as tubes along the route with vertical drops; conductor colours configurable (IEC 60445 defaults).
Export	GLB / glTF (metres, component ID, reference, type and pin nets in node extras), OBJ and STL (geometry only — format limitation).

3D needs for the next stage

- **Manufacturer-accurate models:** import STEP/GLB models from manufacturers for selected parts (licence review needed per vendor).
- **Building context:** IFC import for walls, ceilings and openings; multi-floor display.
- **Performance on large buildings:** instanced rendering for repeated devices; level-of-detail meshes.
- **Hardware:** any GPU with WebGL 2 is sufficient for current scenes ESTIMATE; integrated graphics have not yet been benchmarked.

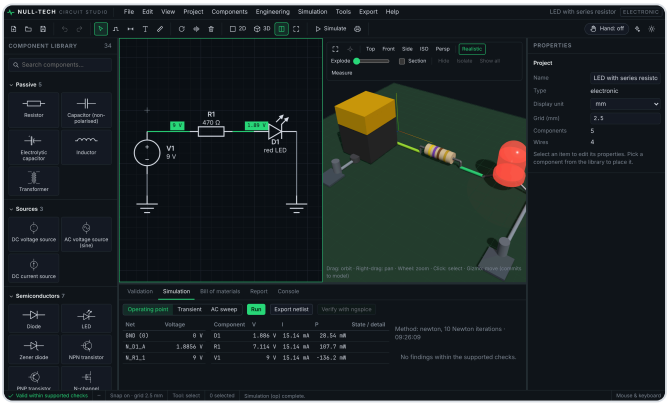


Fig. 7 — 2D and 3D side by side from one model; simulated voltages on the schematic, LED lit in 3D.

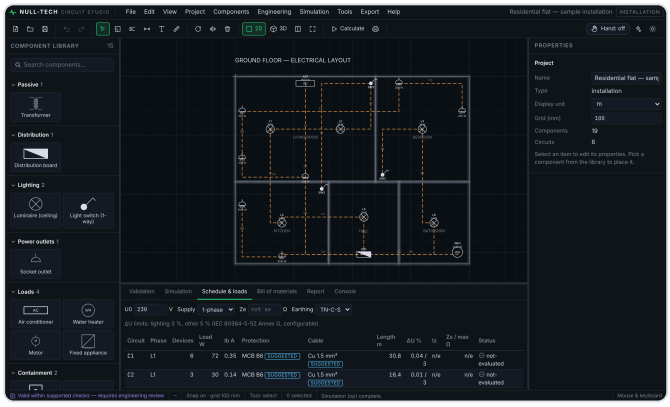


Fig. 8 — Installation plan that generates the 3D building in Fig. 2.

11 Photorealistic visualisation (Blender pipeline)

Beyond the real-time 3D viewport, the same engineering model can be rendered photorealistically for presentations, client reviews, teaching material and marketing. The images on this page and the next are **rendered from the actual project files**, not drawn by hand.



Fig. B1 — Residential flat, rendered with Blender Cycles from the app's GLB export: walls extruded from room polygons, luminaires at 2.7 m, socket and switch plates at mounting height, cable routes at ceiling level.

```
NULL-TECH model ──Export ▸ 3D: GLB──▶ GLB (metres, component IDs and nets in node extras)
      ──▶ tools/blender/render.py  (headless: blender -b -P render.py -- in.glb out.png MODE W H
SAMPLES …)
      ──▶ modes: building · board (circuit close-up with depth of field) · technical (Workbench
outlines)
```

Aspect	Detail
Renderer	Blender 4.0, Cycles path tracer on CPU (no GPU needed); Workbench for technical views
Scene set-up (automatic)	model import, outward normals recomputed, ground or solder-mask plane, sun + area light, neutral world, camera framed on the model bounding box
Fidelity	geometry, materials and simulation state (lit LEDs) come from the model; resistor colour bands are computed from the actual resistance (IEC 60062)
Render time MEASURED	≈ 17 s for an 800 × 500 preview of the flat (48 samples, 4 CPU cores); final 1600–1800 px images ≈ 2–5 min each
Quality improvement found	rendering exposed overlapping walls where rooms share only part of an edge; plus double-covered corner squares where walls meet; the 3D generator was fixed (walls split at every room corner, each junction square owned by exactly one wall) and tests added

Renders of electronic and control projects

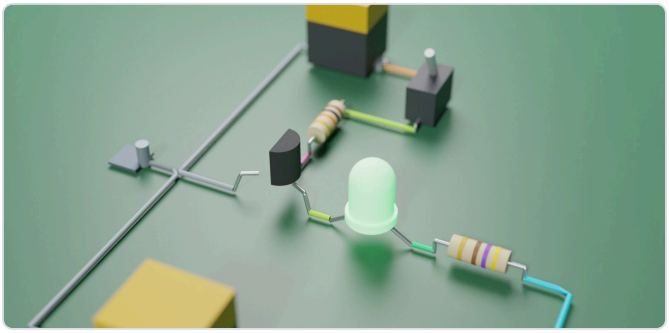


Fig. B2 — Transistor switch: 9 V battery, TO-92 transistor, colour-coded resistors, green LED lit by the simulated collector current.

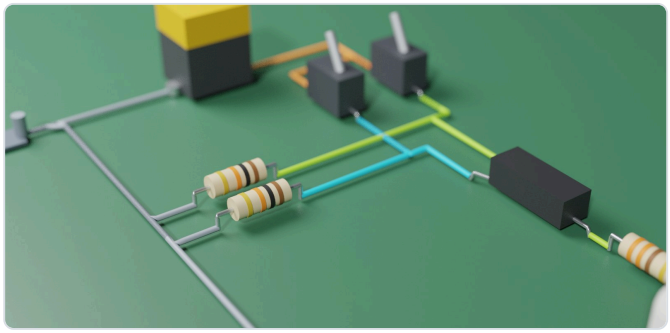


Fig. B3 — Logic-gate circuit: toggle switches with pull-down resistors, DIP gate, LED output.

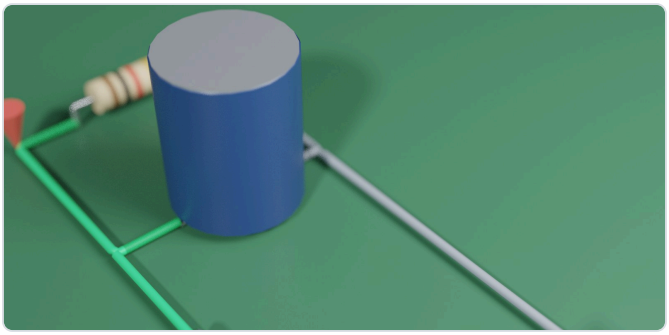


Fig. B4 — Half-wave rectifier: 1N4007 diode, 470 µF electrolytic capacitor (size scales with capacitance), 1 kΩ load.

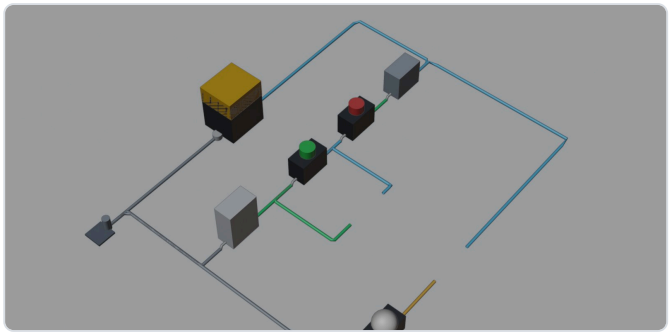


Fig. B5 — Technical (Workbench) view of the motor-starter control circuit: push buttons, contactor coil, overload contact.



Fig. B6 — Low-angle cutaway of the flat: devices and cable drops between ceiling route and mounting heights.

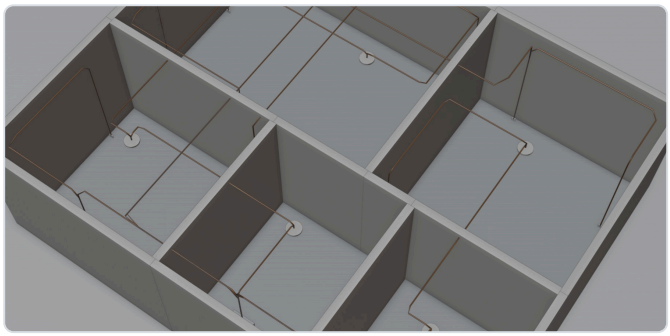


Fig. B7 — Technical outline view for documentation and teaching handouts.

3D production needs

Need	Purpose	Effort / cost
Blender (free, open source)	photoreal renders, animations, marketing images	USD 0
Render workstation	faster renders: any modern GPU cuts minutes to seconds ESTIMATE	optional; CPU works today
Manufacturer 3D models (STEP/GLB)	exact package shapes for popular parts and panels	licence review per vendor
3D artist (part-time)	material library, presentation scenes, turntable animations	≈ 1 person-month in Phase 5 ESTIMATE

12 Data

12.1 Data inventory

Data	Content	Where stored	Personal data?
Project files	Components, wiring, rooms, circuits, settings, revisions	User's computer (`.ntcs.json`)	Author name only, if entered
Autosave & snapshots	Copy of the current project, up to 10 snapshots	Browser storage on the user's computer	As above
Engineering reference data	Resistivities, standard sizes, breaker ratings, ΔU limits, trip multiples — each with its source	Shipped with the software (JSON)	No
User ampacity tables & prices	Values the designer enters, with source	Inside the project file	No
Camera frames	Live video for landmark detection	Memory only; newest frame kept, never written to disk	Yes (image) — never leaves the machine
Hand landmarks	21 points per hand per frame	Memory; optional recording with `--record`	Low (no image)
Shortcuts, favourites	UI preferences	Browser storage	No

12.2 Data flows and privacy

- No account, no telemetry, no cloud upload: the application works fully offline.
- The camera service listens only on 127.0.0.1 and refuses browser connections from non-local origins (tested).
- ngspice verification runs locally; netlists with control blocks, includes or shell commands are rejected (tested).
- Recommendation for institutional roll-out: a short privacy notice explaining camera use, and an explicit on/off control (already present).

12.3 Exchange formats

Format	Direction	Preserves	Limits
.ntcs.json	read / write	everything; versioned, schema-validated	—
PDF / SVG	write	vector drawings at true scale, title block	not re-importable
PNG / JPEG	write	raster drawings (200 dpi)	raster
DXF R12	read / write	geometry, layers, text, units	blocks, hatches, splines not imported
SPICE (.cir)	read / write	R C L V I D Q M, models, connectivity	subcircuits, controlled sources reported
GLB / glTF	write	geometry, materials, component identity	—
OBJ / STL	write	geometry	no identity (format)
CSV / HTML	write	bill of materials; engineering / lab report	—

12.4 Data quality controls

Every file is parsed, migrated and validated on open (unknown component types, broken wire references, missing boards and malformed polygons are reported precisely). All 15 sample projects are validated against the published JSON Schema in the test suite.

13 Security and threat model

The product runs locally with no accounts or servers, which removes most network attack surface. Remaining threats and their treatment:

Threat	Entry point	Mitigation	Status
Malicious or corrupt project file	File ▸ Open	strict parse + schema validation + reference checks before loading; no code execution from files	tested
Web page hijacking the local camera service	WebSocket on 127.0.0.1	binds to loopback only; browser connections refused unless the origin is localhost	tested
Command injection through ngspice	"Verify with ngspice" netlist	.control, .include, .lib and shell lines rejected; runs in a temporary directory with timeout	tested
Camera privacy	webcam frames	frames processed in memory, newest only, never stored or transmitted; explicit enable/disable	by design
Malicious DXF / SPICE import	import menus	parsers only extract geometry / elements; unsupported content reported, never executed	tested
Supply-chain compromise	npm / PyPI / crates	small dependency set, lock files, CI builds; add dependency audit in release gate	planned
Tampered installer	downloads	code-signed installers (Windows, macOS), checksums	Phase 4
Desktop file access misuse	Tauri file APIs	capabilities limited to dialog-selected save/open; no shell access granted	configured

14 Accessibility and localisation

Area	Current state	Next step
Input alternatives	mouse, keyboard (every command has a shortcut, remappable), hand gestures	voice commands (research)
Visual	dark and light themes; status shown by icon + text + colour (not colour alone)	high-contrast theme; scalable UI
Screen readers	labelled toolbar, dialogs and panels	audit with NVDA / VoiceOver
Languages	English; translation layer in place, RTL-ready	Arabic, Kurdish (Sorani) by qualified translators
Regional rules	all limits and tables are project data with sources	Iraq / Kurdistan rule pack after review

15 Hand tracking and human factors

Gesture	Command
Index finger pointing	move cursor
Thumb–index pinch	select / drag / draw
Release pinch	finish operation
Open palm (held)	cancel + pause
Fist (held)	confirm preview
V-sign (held)	reject preview
Two hands pinching	pan / zoom

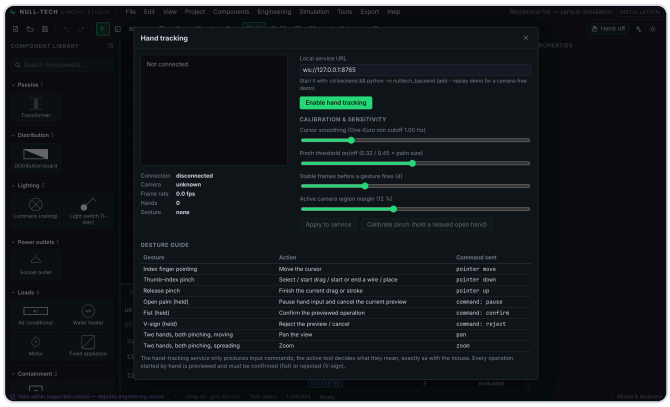
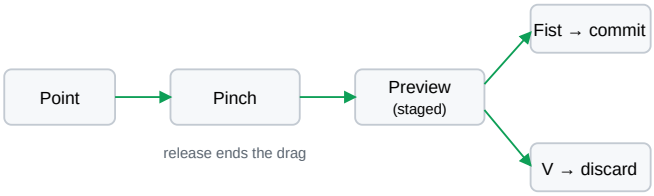


Fig. 9 — Hand-tracking panel: connection, camera status, landmark preview, sensitivity and calibration, gesture guide.



Robustness built in

- Newest-frame capture (no backlog); hand lost during a drag → automatic release after 0.35 s; reconnect with back-off; mouse and keyboard always active.
- Calibration from a relaxed open hand; adjustable smoothing, pinch thresholds, stable-frame count and active camera region.
- Every gesture-initiated change is previewed and requires a confirm gesture — accidental gestures cannot change the design.

Human-factors work still required

Study	Purpose	Measure
Real-webcam latency	End-to-end delay from hand movement to cursor	ms, by camera and CPU class
Recognition accuracy	Different hands, skin tones, lighting, backgrounds	false accept / reject per gesture
Task performance	Place, wire, confirm — hand vs mouse	time, errors, SUS score
Fatigue	“Gorilla-arm” over longer sessions	self-reported effort (Borg scale)
Accessibility	Users with limited fine motor control	task completion

16 Requirements: hardware, software, people

16.1 End-user computer ESTIMATE

Item	Minimum	Recommended
Processor	dual-core, 64-bit	quad-core (hand tracking + 3D at the same time)
Memory	4 GB	8 GB
Graphics	WebGL 2 capable (integrated is fine)	any discrete or recent integrated GPU
Display	1366 × 768	1920 × 1080 or larger; second screen for drawings
Browser / OS	recent Chrome or Edge; Windows 10+, macOS 12+, Linux	desktop app (Tauri)
Webcam (optional)	720p at 30 fps	1080p at 30–60 fps, fixed mount, even lighting
Python (optional)	3.10+ for hand tracking	bundled in the future desktop installer
Disk	a few MB in the browser; desktop installer size not yet measured	+ project files (typically < 1 MB each)

Basis: the main application bundle is ≈ 0.33 MB compressed and calculations take milliseconds (measured, §6). These requirements have not yet been benchmarked on low-end machines; doing so is part of the pilot.

16.2 Development and test environment

Need	Detail
Toolchain	Node.js 20+, TypeScript 5.9, Vite 7, React 19, Three.js; Python 3.10+, OpenCV, MediaPipe, websockets; Rust + Tauri 2 for desktop builds; ngspice for verification.
Continuous integration	Type-check, 88 application tests, 65 service tests, production build, 5 browser workflows on every change (GitHub Actions workflow included).
Test hardware	Windows, macOS and Linux machines; 3–5 webcam models; a low-end laptop; projector for classroom tests.
Signing	Apple Developer Program, Windows code-signing certificate.

16.3 Non-technical needs

- **Licensed electrical engineer** to review every calculation rule and reference value (mandatory before professional use).
- **Standards access:** current editions of IEC 60364 parts 4-41, 4-43, 5-52; IEC 60898-1; IEC 60228; and the applicable national rules.
- **Regional rules:** written requirements for Iraq / Kurdistan Region from the competent authority.
- **Translators** for Arabic and Kurdish (Sorani) with technical vocabulary.
- **Pilot partners:** at least one university department or training centre.
- **Legal:** licence terms, disclaimer wording, privacy notice, third-party licence review (all current dependencies are open source).

17 Team roles and responsibilities

Role	Responsibilities	Involvement
Product owner / project manager	scope, priorities, pilot partners, schedule, budget, reporting	0.5 FTE
Electrical engineer (licensed)	review and sign-off of calculation rules, reference data, regional rules, test cases	0.3–0.5 FTE
Senior software engineer (frontend / 3D)	model, tools, 2D/3D views, drawings, performance	1 FTE
Software engineer (simulation / calculations)	solver, device models, installation engine, verification against references	1 FTE
Computer-vision / Python engineer	hand tracking, calibration, latency and accuracy studies, desktop sidecar	0.5 FTE
QA / test engineer	test plans, regression, cross-platform and webcam testing, accessibility	0.5 FTE
UX / UI designer	usability studies, RTL layout, onboarding, teaching material design	0.3 FTE
Education content author	tutorials, exercises, lab sheets aligned with courses	0.3 FTE (phase 1, 3)
DevOps / release engineer	CI, signed installers, update channel	0.3 FTE (phase 4)
Translators	Arabic, Kurdish (Sorani)	contract

Responsibility matrix (R = responsible, A = accountable, C = consulted, I = informed)

Activity	PO/PM	Elec. eng.	SW frontend	SW sim.	CV eng.	QA	UX
Calculation rules & reference data	I	A	C	R		C	
Simulation accuracy	I	C		A/R		R	
2D/3D/drawings	I	C	A/R	C		R	C
Hand tracking	I		C		A/R	R	C
Localisation & RTL	A	C	R			R	R
Pilot & user studies	A	C	C	C	C	R	R
Release & installers	A		R		R	R	

18 Roadmap and timeline ESTIMATE

Phase	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
1 · Pilot, engineering review, webcam studies	■	■	■									
2 · Safety calculations completion			■	■	■	■						
3 · Localisation & regional rules					■	■	■					
4 · Desktop release (signed installers)							■	■				
5 · Extensions (PLC timers, ladder, IFC, timing)									■	■	■	■

Phase	Main deliverables	Exit criterion
1	pilot deployment; rule review report; webcam latency/accuracy study; usability findings	review sign-off; success criteria in §21 met
2	line-to-line short circuit, selectivity, k^2S^2 thermal withstand, IT systems, grouping/temperature derating with sourced tables	each formula tested against hand calculation and reviewed
3	Arabic and Kurdish interface with RTL; regional rule pack	native-speaker review; rule pack signed off
4	signed Windows/macOS/Linux installers with bundled hand-tracking service; update channel	clean install on all three platforms
5	PLC timers/counters and ladder view; timed digital simulation; conduit routing; IFC exchange; manufacturer 3D models	per-feature tests and documentation

Already completed (version 0.1)

Foundation model, component library, simulator with ngspice validation, installation engine with fault-loop checks, tray routing, logic and PLC simulation, 2D/3D views, drawings and exports, hand-tracking service, shared input system, assistant, examples, documentation, desktop shell, automated test suites.

19 Cost estimate

> **How to read this section.** Effort (person-months) is an engineering estimate ESTIMATE. Monetary values use two **illustrative blended rates** ASSUMPTION — replace them with your local salary or contractor rates; every total is simply effort × rate.

19.1 Development effort by phase and role (person-months)

Role	Ph 1	Ph 2	Ph 3	Ph 4	Ph 5	Total
Product owner / PM	1.5	1.5	1.0	1.0	2.0	7.0
Electrical engineer (review)	1.5	2.0	1.0	0.0	1.0	5.5
Software — frontend / 3D	2.0	2.0	2.0	1.0	4.0	11.0
Software — simulation / calculations	1.0	4.0	1.0	0.0	3.0	9.0
Computer-vision / Python	1.5	0.0	0.0	1.0	0.5	3.0
QA / test	1.5	1.5	1.0	1.0	1.5	6.5
UX / UI	1.0	0.0	1.0	0.0	0.5	2.5
Education content	1.0	0.0	0.5	0.0	0.5	2.0
DevOps / release	0.0	0.0	0.0	1.5	0.0	1.5
Total person-months	**11.0**	**11.0**	**7.5**	**5.5**	**13.0**	**48.0**

19.2 Labour cost at illustrative rates ASSUMPTION

Phase	Person-months	at USD 1 500 / PM	at USD 4 000 / PM
1 · Pilot & review	11.0	16 500	44 000
2 · Safety calculations	11.0	16 500	44 000
3 · Localisation	7.5	11 250	30 000
4 · Release	5.5	8 250	22 000
5 · Extensions	13.0	19 500	52 000
Total	**48.0**	**72 000**	**192 000**

19.3 Non-labour costs

Item	Basis	Indicative amount
Apple Developer Program	required to sign/notarise macOS builds; published price	USD 99 / year
Windows code-signing certificate	varies by provider and validation level ESTIMATE	USD 200–700 / year
Standards documents	IEC 60364 parts, 60898-1, 60228 + national rules; per-document prices vary ESTIMATE	USD 1 000–3 000 once
Test webcams	3–5 models for accuracy studies ESTIMATE	USD 150–500 once
Test computers	low-end laptop + macOS machine if not available ESTIMATE	USD 0–2 500 once
Translation	Arabic + Kurdish UI and documentation; quote required ASSUMPTION	quote
Servers / cloud	none required (local-first); optional website and download hosting	≈ USD 0–20 / month
Open-source licences	all current dependencies are permissive open source	USD 0

Contingency of 15–20 % on labour is recommended for an engineering product of this kind. Phase 1 alone (≈ 11 person-months) is the smallest self-contained step and produces the evidence needed to decide on the remaining phases.

20 Three-year cost of ownership and sensitivity

> Same method as the previous section: effort in person-months is an ESTIMATE ; money uses the illustrative blended rates ASSUMPTION . Maintenance is assumed at 2.5 person-months per year (bug fixes, dependency updates, standards updates, support) ESTIMATE .

Item	Year 1	Year 2	Year 3	3-year total
Development (person-months, phases 1–5)	48.0	—	—	48.0
Maintenance & support (person-months)	—	2.5	2.5	5.0
Engineering rule review (person-months)	incl.	0.5	0.5	1.0
Total person-months	**48.0**	**3.0**	**3.0**	**54.0**
Labour at USD 1 500 / PM	72 000	4 500	4 500	81 000
Labour at USD 4 000 / PM	192 000	12 000	12 000	216 000
Recurring fixed (signing, domain, hosting) ESTIMATE	300–1 050	300–1 050	300–1 050	900–3 150
One-off (standards, webcams, test machines) ESTIMATE	1 150–6 000	—	—	1 150–6 000

Sensitivity of the development budget (48 person-months)

Blended rate → / effort ↓	USD 1 500	USD 2 500	USD 4 000	USD 6 000
−20 % effort (38.4 PM)	57 600	96 000	153 600	230 400
Base (48 PM)	72 000	120 000	192 000	288 000
+20 % effort (57.6 PM)	86 400	144 000	230 400	345 600

Lowest-commitment options

Scope	Person-months	What you get
Phase 1 only (pilot & review)	11.0	evidence from real users and a licensed engineer to decide the rest
Phases 1 + 4 (pilot + release)	16.5	signed installers of today's feature set, reviewed rules
Phases 1–4	35.0	professional-grade release with localisation and full safety calculations

Running cost stays near zero because the product needs no cloud: there are no server, database or per-user infrastructure fees.

21 Pilot plan and success criteria

Element	Plan
Participants	1–2 university departments or training centres; ≈ 30–60 students; 3–5 instructors; 2–3 practising electrical designers ESTIMATE
Duration	12 weeks: 2 weeks set-up and training, 8 weeks use in courses, 2 weeks evaluation
Activities	build the 15 examples manually; one design assignment per mode; installation exercise on a real floor plan; hand-tracking sessions
Engineering review	licensed engineer re-derives every formula and reference value; compares 20 installation circuits against independent calculations
Data collected	task times, errors, usability (SUS) questionnaire, instructor interviews, crash/bug reports, webcam latency and recognition logs (landmarks only, with consent)

Success criteria (proposed targets)

Criterion	Target	Measurement
Calculation correctness	no unresolved discrepancy in the engineering review	review report
Simulation agreement	all examples within 0.1 mV of ngspice (already met)	automated tests
Task completion	≥ 90 % of students complete the LED, divider and installation tasks unaided	observation
Usability	SUS score ≥ 70	questionnaire
Hand tracking	≥ 95 % gesture recognition in normal lighting; cursor latency acceptable to users	logged study
Stability	no data loss; crash-free sessions ≥ 99 %	bug reports, autosave recovery
Instructor value	≥ 3 of 5 instructors willing to use it next term	interviews

Key performance indicators after launch

- Active institutions and users per term; projects created; examples completed.
- Share of designs with zero errors at export; number of findings resolved before printing.
- Reported calculation discrepancies (target: zero open); median time to fix a bug.

22 Editions, go-to-market, support and training

Edition comparison FOR DECISION

Feature	Student (free)	Institutional	Professional
Electronic & control modes, simulation	✓	✓	✓
Examples, tutorials, lab reports	✓	✓ + course packs	✓
Installation mode & calculations	study use	✓	✓
Drawings with custom title blocks	watermark	✓	✓
Regional rule packs (reviewed)	—	✓	✓
Hand tracking	✓	✓	✓
Blender render pipeline	—	✓	✓
Support	community	email + onboarding	priority

Go-to-market sequence

1. **Pilot institutions** (Phase 1): co-develop course packs; collect testimonials and usage data with consent.
2. **Academic launch**: free student edition; institutional licence for labs; workshops for instructors.
3. **Professional launch** (after regional rule review): designers and contractors; reviewed rule packs.
4. **Channels**: engineering faculties, professional syndicates and associations, training centres, technical conferences, online video tutorials.

Support and maintenance plan

Activity	Frequency / target
Bug-fix releases	monthly; critical calculation issues within 72 h of confirmation
Dependency and security updates	quarterly, plus urgent advisories
Standards / rule-pack updates	when a referenced edition changes; reviewed before release
Calculation-discrepancy reports	logged publicly, reproduced with a test, fixed and credited

Training plan

Audience	Format	Duration
Instructors	hands-on workshop: examples, assignments, assessment with reports	1 day
Students	guided tutorials built into the software (15 examples)	self-paced
Designers	installation workflow, calculation transparency, drawing production	½ day
Lab technicians	hand-tracking set-up, camera placement, calibration	2 hours

23 Risks and mitigations

Risk	Likelihood	Impact	Mitigation
A calculation is used as if it were a compliance certificate	Medium	High	no “safe” status; disclaimers on every drawing and report; engineering review; “not evaluated” for missing checks
Error in a formula or reference value	Low–Medium	High	cited sources; hand-calculation tests; licensed review; versioned rule packs
Hand tracking unreliable in real classrooms	Medium	Medium	pilot study; calibration; mouse always available; preview/confirm
Regional rules unavailable or ambiguous	Medium	Medium	rules stored as data; work with competent authority; mark as “review”
Performance on low-end school computers	Medium	Medium	benchmark in pilot; simplified 3D mode; level of detail
Key-person dependency	Medium	Medium	documentation, tests and CI already in place; pair work
Third-party library changes	Low	Low–Medium	pinned versions, CI, small dependency set
Camera privacy concerns	Low	Medium	frames never leave the machine; explicit on/off; privacy notice

24 Quality, safety and governance

- **Status vocabulary:** VALID within supported checks WARNING ERROR NOT EVALUATED REQUIRES REVIEW.
- **Traceability:** every result lists formula, inputs, assumptions and source; reference data carries its citation.
- **Change control:** calculation changes require a test against an independent calculation and engineering sign-off.
- **Release gates:** all automated suites green; ngspice agreement; manual exploratory test on each platform.
- **Physical-safety boundary:** the software is for design assistance and education; installation and machine-safety functions require qualified design, standards compliance and controlled commissioning.

25 Detailed work breakdown

Phase	Work package	Tasks	PM
1	Pilot operations	partner agreements, installation in labs, instructor training, support during term	3.0
	Engineering review	re-derive every formula and reference value; 20-circuit independent comparison; report	2.0
	Hand-tracking studies	latency and accuracy per camera; lighting tests; calibration improvements	2.5
	Usability & fixes	observation, SUS survey, fixes from findings, education content	3.5
2	Fault & short-circuit	line-to-line Isc, IT systems, k^2S^2 thermal withstand, tests vs hand calculation	4.5
	Selectivity & derating	breaker selectivity tables (user-supplied), grouping and ambient factors with sources	4.0
	Review & QA	engineering sign-off, regression suite	2.5
3	Localisation	string extraction, Arabic & Kurdish, RTL layout, fonts, drawings in RTL	4.5
	Regional rule pack	encode reviewed regional requirements as data; tests	3.0
4	Packaging	Tauri installers ×3 OS, Python sidecar, signing, update channel	3.5
	Release QA	clean-install matrix, documentation, release notes	2.0
5	PLC extensions	TON/TOF/CTU, ladder-diagram view of the program	3.5
	Timed digital simulation	event-driven engine, flip-flops, propagation delay	3.5
	BIM / IFC & conduits	IFC import/export of electrical elements; conduit routing	4.0
	3D library & renders	manufacturer models, render presets, animations	2.0
Total			**48.0**

26 Decision requested

1. Approve **Phase 1 (pilot and engineering review)**: ≈ 11 person-months over 3 months.
2. Nominate the pilot institution(s) and the reviewing licensed electrical engineer.
3. Provide the regional regulations and the standards editions to be applied.
4. Choose a business-model option (§9.5) to guide Phases 3–4.
5. Confirm local rates so §19 can be converted from person-months to a final budget.

Appendix A — Standards and references

Reference	Used for
IEC 60364-4-41	fault protection: TN condition Zs·Ia ≤ U0, TT condition RA·IΔn ≤ 50 V, RCD additional protection
IEC 60364-4-43	overcurrent protection Ib ≤ In ≤ Iz; breaking capacity
IEC 60364-5-52 (Annex G, Table 52.2)	voltage-drop method and limits; minimum conductor size
IEC 60898-1	breaker rated currents; instantaneous trip ranges B/C/D
IEC 60228 · IEC 60028	conductor sizes; copper resistivity
IEC 60617 · IEC 60445 · IEC 60062	graphical symbols; conductor colours; resistor colour code
IEC 61131-3 · IEC 61131-2	Structured Text subset; PLC input thresholds
ISO 216 · ISO 5457	paper sizes; drawing frames and zones
Ho, Ruehli & Brennan (1975); Nagel, SPICE2 (1975)	modified nodal analysis; circuit simulation methods
Casiez, Roussel & Vogel (2012)	One Euro filter

Appendix B — Test inventory MEASURED

Suite	Tests	Scope
Engineering (simulation, logic, PLC, installation)	63	hand calculations, ngspice cross-checks, fault loop, routing
Integration (exchange, input system, 3D/2D)	22	formats, mouse = hand, preview/confirm, undo/redo, pin coincidence
Unit (containment routing)	3	tray graph routing
Application total	**88**	all passing
Python service	65	gestures, smoothing, controller, protocol, security, schemas
End-to-end browser workflows	5	real app + real hand-tracking service

Appendix C — Glossary

MNA	Modified Nodal Analysis — matrix method for solving circuits	**Ib / In / Iz**	design current / breaker rating / cable current-carrying capacity
Ze / Zs	external / total earth-fault loop impedance	**ΔU**	voltage drop
RCD / RCBO	residual-current device / combined with overcurrent breaker	**TN / TT / IT**	earthing systems
ngspice	open-source SPICE circuit simulator used as reference	**Structured Text**	PLC programming language of IEC 61131-3
GLB / glTF	standard 3D formats for real-time use	**DXF**	CAD drawing exchange format
Landmarks	21 tracked points on a hand	**SUS**	System Usability Scale (0–100)

NULL-TECH Circuit Studio is engineering assistance software. Designs produced with it must be verified by a qualified engineer against the applicable standards and local regulations before construction or installation. All measured values in this document were produced from version 0.1 of the software on the date of writing.